



NSF Protocol

NSF P248.02-2025

Military Operations Chemical Water Purifiers

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NSF P248.02-2025

NSF Protocol
for Water Systems –

Military Operations Chemical Water Purifiers

Protocol Developer

NSF

Defense Health Agency – Public Health

US Army Combat Capabilities Development Command – Soldier Center

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The following table is provided as a reference for unit abbreviations for common forms of measurement used within NSF documents.

Time	second	s
	minute	min
	hour	h
	day	d
	week	wk
	month	mo
	year	yr
Length	inch	in.
	foot	ft
	yard	yd
	micrometer	µm
	nanometer	nm
	millimeter	mm
	centimeter	cm
	meter	m
	kilometer	km
Liquid measure	milliliter	mL
	liter	L
	liters per day	LPD
	liters per minute	LPM
	ounce	oz
	pint	pt
	quart	qt
	gallon	gal
	gallons per minute	GPM
	gallons per day	GPD
Weight	microgram	µg
	picogram	pg
	nanogram	ng
	milligram	mg
	centigram	cg
	gram	g
	kilogram	kg
	pound	lb
	ton	t
	metric ton	mt

Miscellaneous	inch of mercury	inHg
	kilopascal	kPa
	negative log of hydrogen ion concentration	pH
	pounds per square inch	psi

Acronyms

The following list is provided as a reference for acronyms used within this document.

A2LA	American Association for Laboratory Accreditation	SAL	single agar layer
ANSI	American National Standards Institute	SC	Soldier Center
APHC	US Army Public Health Center	SUWP	small unit water purifier
CONUS	continental United States	SWP	small water purifier
COTS	commercial-off-the-shelf	SWPS	squad water purification system (includes fire team)
DEVCOM	Combat Capabilities Development Command	TDS	total dissolved solids
DOD	Department of Defense	TG	technical guide
DHA-PH	Defense Health Agency – Public Health	TIC	toxic industrial chemicals
DI	deionized	TOC	total organic carbon
ETO	equipment testing organization	TDS	total dissolved solids
GRA	Government Review Agency	US EPA	US Environmental Protection Agency
IAW	in accordance with	UV	ultraviolet
ISO	International Organization for Standardization	UVV	unit void volume
IWP	individual water purifier	VOC	volatile organic compound
MCL	maximum contaminant level	WHO	World Health Organization
MEG	Military Exposure Guideline		
NELAP	National Environmental Laboratory Accreditation Program		
NPDWR	National Primary Drinking Water Regulations		
OCONUS	outside continental United States		
POC	point-of-contact		
PE	performance evaluation		
pH	negative log of hydrogen ion concentration		
PSTP	product-specific test plan		
PWP	platoon water purifiers		
QA	Quality Assurance		
QC	Quality Control		
R&D	Research and Development		
RO	reverse osmosis		

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Foreword

This document, NSF P248.02: *Military Operations Chemical Water Purifiers*, is a follow-on to NSF P248.01: *Military Operations Microbiological Water Purifiers*, which has a scope that is limited to microbiological contaminants (bacteria, virus, protozoan cysts). The US Department of Defense (DOD) set forth to develop NSF P248.02 out of the recognized need to develop a chemical reduction test protocol so that all small water purifiers (SWPs) to be developed by or marketed to the DOD could be evaluated against to determine their effectiveness in providing reduction of targeted chemicals from source water. Devices that successfully completed the test protocol could then be used by deployed personnel with a greater measure of assurance that the water they obtain by using the devices would be of acceptable chemical quality and protective of Warfighter health.

This protocol has been developed with the firm hope that vendors who desire to market their SWPs to the DOD will use it to develop, evaluate, and improve their devices. Additionally, it is recommended for reference in requirements documents, use during government down-selection, and during combat development efforts to determine device capabilities. The ultimate goal is to provide our Warfighters the capability to produce sustainable quantities of chemically safe drinking water from any freshwater source that may be available to them during deployments.

This protocol was developed by the DOD, Defense Health Agency – Public Health (DHA-PH), Defense Centers for Public Health – Aberdeen, Environmental Health Sciences and Engineering Directorate/Field Water Branch and US Army Combat Capabilities Development Command (DEVCOM) – Soldier Center (SC), Individual Hydration Program during FY 2023 to 2025. The development was in coordination with personnel from NSF, who provided guidance and assistance in developing the protocol and offered to publish, maintain, and distribute the document.

DHA-PH and DEVCOM-SC also solicited and received input from many other DOD organizations who contributed significantly to the refinement of this protocol, including the US Army DEVCOM – Ground Vehicle Systems Center, US Army Program Executive Office – Soldier, US Air Force Medical Command, US Navy and Marine Corps Force Health Protection Command, US Navy Naval Facilities Engineering Command, and the US Marine Corps Systems Command whose many efforts of individuals from all of these and other organizations across DOD Services are greatly appreciated. As such, this protocol was finalized through the Joint Medical Field Water Subgroup to the Joint Environmental Surveillance Working Group, Defense Health Agency, as a multi-Service-endorsed testing protocol.

This protocol is intended to provide a framework of procedures for evaluating chemical water purifiers that are of importance to military operations. Testing must follow the prescribed procedures, or others pre-approved as necessary to thoroughly evaluate unique technologies. The developers and secretariats of this recognize the “living” nature of this protocol. As such, the scope of chemicals evaluated may expand with time and improved or alternate methods of analyses may be referenced. Modifying the list of test chemicals is encouraged if mission dictates or as threats to water supplies change, while keeping with the intent and rigor of the protocol.

This is the first edition of the protocol.

This protocol was developed in collaboration with NSF and an external review panel for the Water Program. The review panel membership includes representation of equipment manufacturers, users, and public health regulators.

This protocol and the accompanying text are intended for voluntary use by certifying organizations, regulatory agencies, users, and equipment manufacturers as a basis for providing assurances that adequate public health protection exists for products covered by the scope of this document.

Suggestions for improvement of this protocol are welcome. This protocol is maintained on a continuous maintenance schedule and can be opened for comment at any time. Comments should be sent to military@nsf.org, or c/o NSF, Standards Department, PO Box 130140, Ann Arbor, Michigan 48113-0140, USA.

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NSF Protocol for Water Systems –

Military Operations Chemical Water Purifiers

1 General

Entrepreneurs and manufacturers have developed a number of SWPs intended to reduce the risk of known or unknown chemical contamination in source water of varying physical, chemical and microbiological quality. SWPs may include IWPs, SWPS, and SUWPs/PWPs. Insofar as these devices are intended for military field use, their performance must be thoroughly evaluated to allow informed product selection, and to protect the health of the Warfighters that use them.

This protocol was derived and adapted primarily from existing publications of the NSF. It describes the procedures to be used to test SWPs designed to achieve reduction of chemical contaminants, including organics and metals, from virtually any fresh water source likely to be encountered by military personnel during training exercises and deployments both within CONUS and OCONUS.

Execution of this protocol for acceptance of an SWP for military use will be administered and monitored by a GRA, currently designated as the DHA-PH, who can be contacted at dha.apg.pub.health-a.list.ehed@health.mil, or <ph.health.mil>. Only the GRA, or delegates thereof, may grant “compliance” with the requirements of this protocol.

1.1 Purpose

For the purposes of this protocol, and with reference to their use by deployed military personnel, SWPs intended for the reduction of chemical contaminants (TICs, or chemical warfare agents) are not necessarily effective for microbiological reduction performance. Claims made by vendors concerning microbiological contaminant reduction efficiencies are not tested and will not be verified by following this protocol. Microbiological reduction testing should follow NSF P248.01. SWPs tested using this protocol are intended for use by individual, squad, or unit/platoon level addressed by devices smaller in size than military-fielded bulk water systems. Devices will be tested for the reduction of one or more of the chemical contaminants identified within the scope of this protocol.

1.2 Scope

Application of SWPs intended for use in non-military operations is within the scope of this protocol. Organizations may request assistance from the GRA for non-military applications.

A PSTP will be developed for each product or type of product based on this protocol and the manufacturer’s use instruction and stated claims. The PSTP should be designed to provide a conservative evaluation of the performance of the SWP against the claims made by the manufacturer.

The protocol is intentionally general in nature to be applicable to all SWPs, irrespective of physical construction or technology employed. If users find procedures described in this protocol inappropriate or inapplicable to

adequately test their particular product or technology, they shall describe proposed deviations and document the reasons for modifying or departing from the protocol in the PSTP. Such deviations and departures will be evaluated by the GRA and judged acceptable as long as the level of testing rigor is not decreased, nor the intent of the protocol altered. Test conditions and procedures shall be designed to specifically challenge the particular technology under evaluation.

1.3 Parties and responsibilities

1.3.1 Test sponsors

Test sponsors (to include vendors/manufacturers/non-government and government organizations) whose devices successfully comply with the performance criteria of this protocol will receive a GRA P248.02 Letter of Compliance stating that the device meets the military requirements of NSF P248.02: *Military Operations Chemical Water Purifiers* for the specific chemical(s) tested. Without the Letter of Compliance, test sponsors shall not claim or market their device as being compliant with this protocol. Information on requirements for GRA compliance can be found in Section [7](#) of this document.

The test sponsor is responsible for:

- notifying the GRA of the intent to test equipment using this protocol
- developing the PSTP (with assistance from the ETO)
- submitting the PSTP to the GRA for review and comment
- selecting, negotiating with, and obtaining the concurrence of the GRA for the third-party ETO
- providing new, commercially packaged purifiers to the ETO for testing, complete with instructions
- providing all necessary resources, including monetary, logistical, and technical support to the ETO to enable testing to proceed on schedule
- ensuring that the final report of testing, in its entirety, is forwarded to the GRA (it is expected that test results will be provided to the GRA regardless of device performance so that the GRA can inform potential users of the health benefits/risks of using the device).

1.3.2 Government Review Agency

The GRA is responsible for:

- assisting test sponsors and ETOs in interpreting and complying with this protocol
- assisting test sponsors, if necessary, in identifying and selecting qualified ETOs
- reviewing and approving PSTPs
- reviewing the laboratory test results and providing a Letter of Compliance to the test sponsor explaining the basis for determining compliance or non-compliance with protocol requirements
- coordinating with the appropriate military offices/officials to distribute the results of testing throughout the US Department of Defense (DOD), as appropriate.

1.3.3 Equipment Testing Organizations

To promote consistent, accurate, and comparable results across laboratories, all laboratories performing testing of water treatment equipment for US Military application under NSF P248.02 are required to

possess/obtain accreditation based on the current version of ISO/IEC 17025,¹ by a recognized third-party accrediting organization (e.g. A2LA,² NELAP,³ etc.). “Scope of Accreditation” for the laboratory must list methods and capabilities equivalent or closely related to the methods and procedures conducted during protocol execution for reporting. All analytical procedures conducted during protocol execution for reporting must be listed on the laboratory’s “Scope of Accreditation” except for those methods that are of “non-routine” nature. These requirements also apply to sub-contract laboratories conducting analyses on behalf of the primary testing laboratory. If the contract laboratories are not accredited to ISO/IEC 17025, the laboratories must demonstrate ISO/IEC 17025 compliance through internal audits by the primary testing laboratory. ISO/IEC 17025 is an internationally accepted standard for developing laboratory management systems for quality, administrative, and technical operations and an accreditation is a crucial requirement for acceptance of test data by the US Military.

The ETO is responsible for:

- providing documentation on accreditation to ISO/IEC 17025, IAW this section
- assisting the test sponsor in developing the PSTP and having it approved by the GRA
- coordinating with the GRA to resolve product- and technology-driven modifications to the protocol
- conducting the testing IAW the GRA-reviewed and approved PSTP
- managing, documenting, evaluating, interpreting, and reporting all test results
- providing a written report documenting the test procedures, deviations from this protocol, results of testing, and the performance evaluation of the SWP as directed in the PSTP
- providing quality assurance and documentation as requested by the GRA.

1.4 Evaluation steps

The testing of SWPs performance involves five sequential steps:

- 1) Development, submission to GRA, and approval of the PSTP.
- 2) Execution of third-party testing IAW the PSTP.
- 3) Data reduction and analysis.
- 4) Report preparation and information transfer.
- 5) Performance review by GRA and issuance of a Letter of Compliance (if applicable).

1.5 Product-specific test plan development, review, and approval

1.5.1 The ETO and the test sponsor must adhere to the requirements of this protocol in developing the PSTP. The final draft PSTP submitted to the GRA for review and approval shall:

- include the information requested in this protocol
- conform to the format identified herein
- conform to the testing procedures described herein

¹ International Electrotechnical Commission. 3 rue de Varamb , 1st floor, P.O. Box 131, CH-1211 Geneva 20, Switzerland. <iec.ch>

² A2LA (formerly American Association for Laboratory Accreditation). 5202 Presidents Court, Suite 220, Frederick, MD 21703. <a2la.org>

³ The NELAC Institute. P.O. Box 2439, Weatherford, TX 76086. <nelac-institute.org>

- include any unique requirements identified by the GRA during initial coordination.

1.5.2 Specifically, the PSTP shall include the following essential elements (see Annex [N-1](#) for a PSTP template):

- the test sponsor's name and address, and a POC name, email address, and telephone number
- the name, address, email address, and phone number of the ETO, and a list of designations, recognitions, certifications, etc., that qualify the ETO to perform the testing
- the names and model numbers of the tested IWP's
- the manufacturer's operating instructions (as an attachment)
- a description of the roles and responsibilities of the testing participants
- a description of the SWP, the basic technologies it uses, and its advertised capabilities
- a list of materials for all parts that contact water, to include certifications as acceptance of materials for contact with drinking water or sufficiently detailed material specifications so that a toxicological analysis can be performed
- the experimental design for the actual testing of the SWP, including any variations from the protocol requirements
- a description of the procedures governing testing activities such as equipment operation and process monitoring; sample collection, preservation, and analysis; and data collection and interpretation.

Once the ETO and the test sponsor have an agreed-upon draft PSTP, it shall be submitted to the GRA for technical review and approval prior to initiation of any testing.

2 Normative references

The following documents contain requirements that, by reference in this text, constitute requirements of this protocol. At the time of publication, the indicated editions were valid. All of the documents are subject to revision and parties are encouraged to investigate the possibility of applying the recent editions of the documents indicated below. The most recent published edition of the document shall be used for undated references.

APHA/AWWA/WEF, *Standard Methods for the Examination of Water and Wastewater* (hereinafter referred to as *Standard Methods*), 21st edition ⁴

ISO/IEC 17025:2017, *General requirements for the competence of testing and calibration laboratories* ¹

NSF P248.01, *Military Operations Microbiological Water Purifiers*

NSF/ANSI 53, *Drinking Water Treatment Units – Health Effects*

NSF/ANSI 58, *Reverse Osmosis Drinking Water Systems*

US EPA, *Guide Standard and Protocol for Testing Microbiological Water Purifiers* ⁵

⁴ American Public Health Association, American Water Works Association, and Water Environment Federation. <[standardmethods.org](https://www.standardmethods.org)>

⁵ US Environmental Protection Agency. 1200 Pennsylvania Avenue NW, Washington, DC 20004. <[epa.gov](https://www.epa.gov)>

US Army Public Health Center, Technical Guide 230: *Environmental Health Risk Assessment and Chemical Exposure Guidelines for Deployed Military Personnel* ⁶

3 Definitions

Terms used in this protocol that have a specific technical meaning are defined here.

equipment testing organization (ETO): An appropriately accredited organization qualified to conduct independent studies and testing of [SWPs](#) IAW protocols and test plans. The role of the ETO is to assist with developing the [PSTP](#) in collaboration with the [test sponsor](#), conduct all performance tests, and [report](#) results against the requirements of the protocol.

individual water purifier (IWP): A device capable of reducing waterborne contaminants that is designed to be carried and used by an individual.

protocol: A written document of procedures for testing that applies to all makes and models of [SWPs](#).

product-specific test plan (PSTP): A written document of procedures for performance and reliability testing that apply to a specific make and model of [SWP](#).

report: A written document that includes data, test results, findings, and any pertinent information collected IAW the protocol and the [PSTP](#).

small unit water purifier (SUWP): A system capable of reducing waterborne contaminants that is designed to be used by a group (unit, platoon, team) of military personnel.

small water purifier (SWP): A generic term used to describe a device/system that is designed to reduce waterborne contaminants and smaller in size than the bulk water systems currently fielded by the US Military.

squad water purification system (SWPS): A system capable of reducing waterborne contaminants that is designed to be used by a squad or fire team.

test sponsor: A manufacturer, vendor, non-government, or government organization wanting to test [SWPs](#). The role of the test sponsor is to supply devices/systems, fund, and collaborate with the [ETO](#) in preparation and execution of the [PSTP](#).

4 Performance requirements

4.1 Chemical reduction requirements

An SWP must be able to demonstrate the chemical reduction capabilities of one or more chemicals shown in Table [4.1](#) when tested to an approved PSTP developed IAW this protocol as described in Section [5](#). Each effluent sample for each unit must continuously meet or exceed the identified reduction requirement of Table [4.1](#), with the exception that up to 20% of effluent samples for each unit may vary from the required reduction provided the average of all samples by unit meet the criteria.

⁶ Defense Health Agency – Public Health. Defense Centers for Public Health – Aberdeen, 8300 Ricketts Point Road, Building E-2850, Aberdeen Proving Ground, MD 21010. <phc.amedd.army.mil>

Table 4.1 provides each chemical target influent challenge concentration and minimum required percent reduction. Chloroform shall be used as a surrogate for VOCs for systems that use activated carbon filter components to accomplish the organic chemical reduction. In validation studies completed by NSF, 0.3 mg/L influent challenge of chloroform was demonstrated to be adequately protective against 0.1 mg/L concentration (contamination) for a select VOC listed as covered by the surrogate. The VOC influent levels of the study varied, informed by occurrence data when available, and the actual average influent levels can be seen in the claims table of NSF/ANSI 53, Table 7.6: *Organic chemicals included by surrogate testing* and NSF/ANSI 58, Section 7.1.1: *Volatile organic chemical reduction claims*. Table 4.2 provides the analytical methods and reporting limits.

Table 4.1**Challenge concentrations and reduction requirements**

Class/Group	Chemical (CASRN)	Target influent challenge concentration (mg/L $\pm$ 20%) ^a	Minimum required percent reduction
inorganic	arsenic V (pentavalent) (7778-43-0)	0.6	$\geq$ 90%
inorganic	arsenic III (trivalent) (7778-43-0)	0.6	$\geq$ 90%
inorganic	cyanide ^b (143-33-9)	2	$\geq$ 90%
inorganic	mercury ^c (7487-84-7)	0.02	$\geq$ 90%
VOC surrogate ^d	chloroform (67-66-3)	0.3	$\geq$ 95%
gasoline/diesel ^e	benzene (71-43-2)	0.05	$\geq$ 90%
	toluene (108-88-3)	5	$\geq$ 90%
	1,3,5-trimethylbenzene (108-67-8)	0.06	$\geq$ 90%
	naphthalene (91-20-3)	0.1	$\geq$ 90%
carbamate	aldicarb (116-06-3)	0.14	$\geq$ 90%
organophosphate	methamidophos (10265-92-6)	0.04	$\geq$ 90%
herbicide	paraquat dichloride (1910-42-5)	0.467	$\geq$ 90%
radionuclide	uranium ^f (7440-61-1)	0.3	$\geq$ 90%
dissolved solid	total dissolved solids ^g (7647-14-5)	see footnote g	N/A ^g

^a Refer to Section 5.7.3.1. Refers to the chemical concentration, not including the salt, where applicable.

^b Added as sodium cyanide.

^c Adsorptive technologies shall be evaluated for mercury reduction performance at pH's 6.5 and 8.5. RO technologies shall be evaluated at pH 7.5.

^d Systems that incorporate activated carbon for the reduction of VOCs may be tested against chloroform IAW NSF/ANSI 53/ NSF/ANSI 58 and claims made based on the appropriate standard.

^e Representative gasoline and diesel chemicals selected due to percent composition, solubility, and toxicity. Toluene represents the organic load contributed by other fuel components based on occurrence data. All listed components tested together.

^f Using uranyl nitrate hexa-hydrate, a depleted uranium.

^g All membrane systems capable of reducing dissolved solids will be evaluated for TDS (as NaCl) reduction IAW Section 5.6.2.2.1. For informational purposes only; no pass/fail criteria.

Table 4.2
Challenge chemicals analytical methods and reporting limits

Chemical	Analytical Method	Reporting limit (mg/L)
chloroform	EPA 524.2	0.0005
cyanide	EPA 335.4	0.005
methamidophos ^a	LC/MS	0.0007
mercury	EPA 200.8	0.0002
benzene	EPA 524.2	0.0005
toluene	EPA 524.2	0.0005
1,3,5-trimethylbenzene	EPA 524.2	0.0005
naphthalene	EPA 524.2	0.0005
paraquat dichloride	EPA 549.2	0.001
arsenic V (pentavalent) ^b	EPA 200.8	0.001
arsenic III (trivalent) ^b	EPA 200.8	0.001
aldicarb	EPA 531.2	0.001
uranium	EPA 200.8	0.001
total dissolved solids (calculated)	EPA 120.1	2

^a No EPA Method is available.

^b Measured as Total Arsenic. Required to verify arsenic speciation of ionic form.

Devices that incorporate uncommon or multiple technologies for chemical reduction may require different procedures than described in this protocol to fully demonstrate device capabilities. Unless it has been shown that a listed surrogate is an acceptable chemical alternative for the given technology, the use of the actual contaminant is required. The PSTP will provide a forum for the manufacturer to provide specific testing procedures for these technologies to the GRA for approval. NSF/ANSI 53, NSF/ANSI 58, or other recognized national standards may be consulted for the selection of appropriate surrogates or procedures to conduct testing.

4.2 Operational capability

Complete systems shall be watertight and shall not leak during performance testing operation.

4.2.1 While this protocol does not presume to dictate the physical characteristics of SWPs that make them useful to or desirable by Warfighters – those characteristics are defined by units, unit commanders, and missions – there are desirable characteristics and operational capabilities that will be considered in the final recommendation as to device suitability for use during military operations.

4.2.2 The characteristics listed below have been selected from current and draft operational requirements documents, and are considered desirable, if not required by the US Military. Commanders planning to purchase specific SWPs will consider these or similar criteria and how they relate to the mission for which the SWPs are to be selected in their decisions. Not all of the following are evaluated by this protocol. The ideal SWP would, in no particular order of importance:

- allow filling and servicing without contaminating the purified water or surfaces that contact the purified water

- be able to be worn with, attached to, or carried on or in current and future load bearing equipment
- be capable of operation by Warfighters in any terrain type and climatic condition (Basic and Hot)
- be fully operational after freeze/thaw cycles
- be compatible with standard military-approved personal hydration systems
- be easily cleaned and disinfected
- be operable by military personnel of all sizes, strengths, and physical conditions
- have a shelf life of 5 yr
- have a treat-to-drink time of less than 20 min
- provide an explicit indication or assurance of the unit's effective use lifetime (in terms of volume purified) to warn the consumer of potential diminished treatment capability through one or more of the following:
 - cessation of water flow
 - a visual change
 - simple, explicit instructions for servicing and replacing components within the recommended use life (measurable in terms of volume throughput, specific time frame, or other appropriate method).
- require little or no training to operate and maintain
- resist mold, fungus, and biofilm growth
- use readily available and easily installed replacement filters and parts.

Additional characteristics specific to IWP:

- have a functional life after initial use of at least 120 d
- have a minimum flow of 200 mL/min at 3 psi vacuum (in-line IWP only)
- treat at least 135 L of water as packaged.

5 Evaluation procedure

5.1 Purpose

The candidate SWP is tested to substantiate its chemical reduction capabilities over the test capacity as defined in Table [4.1](#) using the analytical methods of Table [4.2](#).

5.2 Apparatus

Two production units of each candidate SWP will be tested simultaneously. The units should be new and removed from the commercial packaging just prior to beginning testing or, if appropriate, shall be "reset" with new, production components to perform as new systems in the existing commercial configuration. Manufacturers are welcome to test R&D or "Beta" devices against this protocol; however, GRA involvement, including PSTP and subsequent laboratory data review, will be at the GRA's discretion and protocol compliance will not be granted. This does not apply to government-directed testing of R&D devices.

5.3 Test water properties

Test water shall be prepared using NSF/ANSI 53 or NSF/ANSI 58 test waters, unless deemed inappropriate due to type of technology or interactions with the test chemicals. No disinfectant of any kind shall be present in any of the test waters prior to treatment.

5.4 Product-specific test plan

5.4.1 The following requirements apply to testing of all SWPs. Additional requirements specific to IWPs and SUWPs are provided in subsequent sections. SWPs designed for squads and fire teams will follow either IWP or SUWP requirements based on device configuration.

5.4.2 The test sponsor, together with the ETO, will produce a written PSTP and forward it to the GRA for review and comment prior to beginning testing. It is preferred that the PSTP template provided in Annex [N-1](#) be used to expedite GRA review and be submitted in Microsoft Word to allow for GRA commenting. Specific questions and PSTP submissions should be directed to the Defense Health Agency, Environmental Health Sciences and Engineering Directorate/Field Water Branch.⁷

5.4.3 The ETO will maintain the specified test water quality throughout the test, not just at sample points. The laboratory will verify the test water composition for each test water batch prepared daily to ensure compliance with Section [5.3](#). Water quality compliance with Section [5.3](#) is also to be verified at the end of daily testing unless otherwise specified and approved in the PSTP. Details on how the test water was prepared and transferred to the devices for processing shall be included in the final report.

5.4.4 Testing procedures must replicate field use.

5.4.5 In general, devices should be tested without any prior conditioning or seasoning with test water or any other water. Should conditioning be required the PSTP must clearly explain the procedures, including conditioning water quality, and conditioning steps must be provided to the user in the supplied device instructions. Conditioning steps must be executable in field environments. Conditioning volume does not accumulate towards the test volume.

5.4.6 If the SWP can be operated in more than one way, all methods of operation should be evaluated. Alternatively, if one method can be demonstrated to be the most rigorous of the method options, only that method need be used during testing.

5.4.7 If the SWP uses a cleanable element with a finite number of cleanings, testing must demonstrate that the device meets the chemical reduction requirements after the element has been cleaned the maximum number of times allowed in the manufacturer's directions (if applicable).

5.4.8 Influent chemical concentrations must be maintained IAW Section [5.7.3.1](#) and the specified influent concentrations of Table [4.1](#). Proper procedures must be in place to ensure a homogeneous test solution.

5.4.9 The performance of adsorptive chemical reduction technologies varies based on the number of active adsorptive sites and on reaction kinetics (speed of the adsorptive chemical reaction). It is highly recommended that testing is not terminated until at least three consecutive failing sample points are demonstrated. As adsorptive sites become less available, testing to three consecutive failing sample points verifies the end of device life. For devices where the kinetics are insufficient to meet reduction requirements, testing to three consecutive failing sample points (or longer) allows for the demonstration of device capabilities that may not meet protocol requirements but can help direct device redesign and may still have value to the US Military.

⁷ <dha.apg.pub-health-a.list.ehed@health.mil>

Conditions triggering end of test shall be described in the PSTP.

5.5 Product-specific test plan – Individual water purifier

5.5.1 Operation of devices may be as batch or continuous flow mode IAW the instructions contained in the commercial package. For batches where effluent samples are collected, the entire contents of the batch shall be collected, and sub-samples analyzed.

If continuous-flow testing is preferred by the test sponsor and ETO, the reasoning, proposed plumbing, and test procedures must be specified in the PSTP.

Laboratory evaluations are typically 4.5 d in duration. Two replicate devices will be tested for reduction of each chemical challenge or, when feasible, two or more chemicals can be evaluated in a combined challenge. Each replicate device will be subjected to only one test. For each test a minimum of 135 L per replicate shall be processed, using a daily test volume of 30 L per replicate processed per day for 4 d and 15 L on Day 5. Total test volumes will be adjusted based on device capacity but should not be less than 135 L, unless specified by a military end user or requirements document.

5.5.1.1 Mouth-drawn operation

Products designed such that the only method to generate treated water for consumption is by drawing from the unit by the user's mouth (by creation of a vacuum) shall be evaluated using the test setup of NSF/ANSI 53, Annex N-3: *Test method for evaluating mouth-drawn water treatment units*, Section N-3.2.1: *Set-up* with the exception that the test rig configuration shall be capable to maintain a flow rate of at least 200 mL/min at -3 psi (20.7 kPa vacuum, 6 inHg).

The flow rate shall be maintained at 200 mL/min up to a maximum of -3 psi. If the vacuum to maintain this flow is greater than -3 psi, the flow rate shall be reduced to maintain a vacuum of -3 psi. If flow cannot be maintained at or above 50 mL/min at -3 psi, the test will be stopped.

5.5.1.2 Squeeze bottle operation

Products designed to have a single outlet and a flexible liquid-holding bottle in which the contents can be forced out by squeezing shall be operated according to NSF/ANSI 53, Annex N-4: *Test method for evaluating squeeze bottle water treatment units*, Section N-4.2: *Method – Mechanical gripper apparatus* or Section N-4.3: *Alternate method – Pressurized bottle*.

5.5.2 Batch processing and sampling procedures

Influent and effluent samples shall be collected at start up, and every 10% of capacity. A single influent sample shall be collected to represent the water applied to all replicate devices.

Two example basic sampling schedules are shown in Tables [5.1](#) and [5.2](#). Test plans for a specific model may require variations from the plan shown and should be designed to provide realistic worst-case testing.

Table 5.1**IWP sampling schedule (continuous flow)**

Daily procedure	Day 1	Day 2	Day 3	Day 4	Day 5
1	collect 0% sample (first 1-L draw)	collect 30% sample	collect 50% sample	collect 70% sample	collect 90% sample (first 1-L draw)
2	collect 10% sample	collect 40% sample	collect 60% sample	collect 80% sample	collect 100% sample
3	collect 20% sample	process water to 45% capacity (no sample)	process water to 65% capacity (no sample)	process water to 90% capacity (no sample)	—

Note. Between each sample point, test water will be processed to reach shown percent capacity. Samples will be collected as a minimum 1-L volume, then aliquots drawn for analysis.

Table 5.2**IWP sampling schedule (example 135-L batch flow)**

Daily batch #	Day 1	Day 2	Day 3	Day 4	Day 5
1	0% sample 1 to 3 L	no sample 31 to 33 L	no sample 61 to 63 L	no sample 91 to 93 L	90% sample 121 to 123 L
2	no sample 4 to 6 L	no sample 34 to 36 L	no sample 64 to 66 L	70% sample 94 to 96 L	no sample 124 to 126 L
3	no sample 7 to 9 L	no sample 37 to 39 L	50% sample 67 to 69 L	no sample 97 to 99 L	no sample 127 to 129 L
4	no sample 10 to 12 L	30% sample 40 to 42 L	no sample 70 to 72 L	no sample 100 to 102 L	no sample 130 to 132 L
5	10% sample 13 to 15 L	no sample 43 to 45 L	no sample 73 to 75 L	no sample 103 to 105 L	100% sample 133 to 135 L
6	no sample 16 to 18 L	no sample 46 to 48 L	no sample 76 to 78 L	80% sample 106 to 108 L	—
7	no sample 19 to 21 L	no sample 49 to 51 L	60% sample 79 to 81 L	no sample 109 to 111 L	—
8	no sample 22 to 24 L	40% sample 52 to 54 L	no sample 82 to 84 L	no sample 112 to 114 L	—
9	20% sample 25 to 27 L	no sample 55 to 57 L	no sample 85 to 87 L	no sample 115 to 117 L	—
10	no sample 28 to 30 L	no sample 58 to 60 L	no sample 88 to 90 L	no sample 118 to 120 L	—

Note. Applies to batch flow testing, represented here as 3-L batches. Samples shall be collected as entire batch, then aliquots drawn for analysis.

5.6 Product-specific test plan – Small unit water purifiers

5.6.1 Operation of devices shall be IAW the instructions contained in the commercial package. The plumbing and recommended locations of sampling should be specified in the PSTP.

5.6.2 Testing shall be designed to provide the worst-case scenario. The operation, test duration, and sampling schedule will be dependent on technology type, including, but may not be limited to, adsorption, membrane filtration (including RO), and systems that employ multiple technologies as a treatment train.

5.6.2.1 Physiochemical technologies (e.g. activated carbon) shall be evaluated to the manufacturer's stated treatment capacity or until performance failure (if capacity is unknown), operated continuously for 6 h/d, excluding start-up procedures, and have a rest period of 12 to 18 h/d. Influent and effluent samples shall be collected at the start of testing then every 10% of the tested capacity. For test Day 1, and for each day where a sample collection occurs at the start of a test day, sample collection will take place after the passage of 10 UVVs) (see Section [5.6.2.2.1.c.2-3](#) to determine void volume). Influent and effluent samples will be collected at each sampling point as a minimum 1-L volume then aliquots drawn for analysis.

5.6.2.2 Water systems using only membrane technologies to reduce the chemical challenge will be evaluated as follows.

5.6.2.2.1 Conditioning, membrane performance verification, and void volume determination

- a) On test Day 1, operate the system on NSF/ANSI 58 TDS test water (sodium chloride, NaCl) until TDS reduction has stabilized or a minimum of 1 h, whichever is longer. Continuously monitor TDS to show stabilization.
- b) Consult the membrane specification sheet to determine the conditions used (e.g. temperature, water quality, recovery, pressure) to demonstrate manufacturer-claimed percent TDS reduction. Using those conditions, operate the system until the TDS reduction has stabilized. Continuously monitor TDS to show stabilization. Report stabilized percent TDS reduction in comparison to manufacturer claim and if not within 1% then contact the test sponsor before proceeding.

Note 1. Refer to the manufacturer membrane specification sheet to define stabilized TDS reduction, and if not defined use less than 4% change over 5 min.

Note 2. If the membrane specification sheet only defines test water TDS concentration, then use NSF/ANSI 58 TDS test water with TDS concentration set at the manufacturer specification.

Note 3. Some membranes may use a challenge other than NaCl to verify performance. In those cases, use the manufacturer specified challenge.

Note 4. If possible, operate the system at the membrane specification recovery and pressure during this performance verification.

- c) The void volume must be determined to ensure that the system is fully flushed and that the observed reduction is based on the influent water quality. Due to mixing within the system, the void volume is multiplied by 10 (10 UVV) and that volume is used at test water changes. Determine the 10 UVV using one of the following procedures.
 - 1) Without draining the system after step [5.6.2.2.1.b](#), change the source water to RO/DI then continuously monitor the influent and effluent TDS until the values stabilize to within 4% of each other. Record the time to stabilization. Use 2× the volume produced during the time to stabilization as the 10 UVV.
 - 2) Calculate the void volume of the flow path using the physical dimensions then multiply by 10.
 - 3) Completely fill then drain the system, measure the water volume, then multiply by 10.

5.6.2.2.2 Chemical challenge procedures

- a) Operate the system on test water to produce the 10 UVV, or 1 h, whichever is greater.
- b) Collect 1 L influent and 1 L effluent samples and analyze for chemical, TDS, and pH.
- c) Operate the system for 1 h.
- d) Repeat steps b) and c) to collect 5 more samples. End of test.

5.6.2.2.3 If the system employs a treatment train consisting of multiple technologies, it may be evaluated as a “black box” IAW Section [5.6.2.1](#), or:

- in a manner consistent with NSF/ANSI 58, Annex N-2: *Evaluation methods for systems with multiple technologies – Treatment train*. In all cases, the procedures in Section [5.6.2.2.1](#) must be performed prior to chemical challenge testing
- all multiple technology systems must have the water quality sampled and analyzed after each technology step that can improve the chemical water quality.

5.7 Analytical procedures

5.7.1 General

5.7.1.1 Chemical (influent and effluent) sampling and analysis shall be conducted at each test point specified in the approved PSTP. All physical and chemical analyses shall be conducted IAW procedures in the current version of *Standard Methods*,⁴ or equivalent.

5.7.1.2 If there is significant change the GRA is to be notified prior to continuing the test.

5.7.1.3 In addition to the basic sampling plan, when the devices being tested contain silver, two new devices will be tested using the silver leaching test water and procedures identified in the US EPA *Guide Standard and Protocol for Testing Microbiological Water Purifiers*.⁵ Testing will follow the same testing procedures to ensure the silver is not leached into the effluent at concentrations exceeding the short-term MEG for silver specified in the current edition of the APHC TG 230,⁶ or the corresponding US EPA NPDWR MCL if there is no MEG. If a different bacteriostatic material is used within the device, procedures shall be developed to evaluate leaching of that material.

5.7.2 Analytical methods

All analyses shall be conducted IAW the applicable methods referenced in Table [4.2](#).

5.7.3 Records

All pertinent procedures and data shall be recorded in a standard format and retained for possible review until the report of results has been completely accepted by review authorities, but in no case for less than 1 yr.

5.7.3.1 Acceptance of records

Average influent must be $\pm 20\%$ of the targeted influent with a tolerance of $\pm 30\%$ for any single sample point.

5.8 Scaling

If a manufacturer produces more than one device using identical technology with similar design and construction, it may be appropriate to allow for acceptance of the performance of one device size as representative of the performance of other devices sizes. In this case the GRA will make the determination as to the appropriateness of accepting data for protocol compliance. In all cases the general rule will allow for acceptance when scaling up to three times and down to one-third of the tested device.

6 Product literature

6.1 Instructions

Information setting forth complete, detailed instructions for installation, operation, and maintenance shall be provided with each system. Specific information shall include:

- complete name, address, and telephone number of manufacturer
- model number and trade designation
- flushing and conditioning procedures
- operation and maintenance requirements (including user responsibility, parts, and service).

6.2 Packaging

Replacement components packaging shall be labeled with the following information (where appropriate):

- model number and name of component
- model number of system(s) in which the component is to be used
- name and address of manufacturer
- rated capacity/rated service life
- disinfection and cleaning instructions
- component rated capacity/rated service life
- operating or exchange steps
- required replacement intervals of ultraviolet lamp(s) IAW the manufacturer's instructions (UV systems).

6.3 Performance data sheet

A performance data sheet shall be available to potential buyers for each system and shall include the following information:

- complete name, address, and telephone number of manufacturer
- model number and trade designation
- statement that the system conforms to NSF P248.02 (if applicable), specifying capacity and contaminants tested
- statement that the system is not intended to convert wastewater or raw sewage into drinking water

- rated capacity/rated service life in liters or gallons. Each unique model designation shall not claim a capacity or service life greater than the least reduction capacity or service life that has been verified through testing to NSF P248.02 for the chemical contaminant evaluated
- general operation and maintenance requirements including, but not limited to:
 - suggested frequency of component change or service to system
 - user responsibility
 - statement that while testing was performed under standard laboratory conditions, actual performance may vary.
- model number of replacement component
- electrical requirements
- explanation of how the performance indicator functions, if present.

7 Government review agency compliance

7.1 GRA NSF P248.02 Compliance

7.1.1 The GRA reserves the right to determine compliance with this protocol for military purposes. Only manufacturers whose devices have been tested, whose data have been reviewed by the GRA, and who have received a GRA P248.02 Letter of Compliance, may market those devices as having met the GRA requirements of this protocol. The GRA may only grant compliance with NSF P248.02. The GRA does not review for, and may not grant the use of, the NSF mark, which may only be granted by NSF.

7.1.2 The GRA's involvement with this protocol is to ensure that SWPs that may be used by the US Military undergo a thorough evaluation of device capabilities with respect to reducing the potential for ingestion of identified chemical contaminants. The performance requirements, as described in Section 5, are deliberately challenging to demonstrate device capabilities under stressed conditions and discriminate between the treatment capabilities different SWPs provide for each tested chemical. The goal is to provide the US Military with effective treatment options regardless of whether the devices are provided to Warfighters by the US Military or are produced and marketed as COTS items.

7.1.3 The following describes the minimum requirements that must be met for the GRA to grant NSF P248.02 compliance:

- the PSTP must be reviewed and accepted by the GRA prior to testing initiation
- testing must be conducted by a GRA-approved third-party laboratory to the requirements set forth in the approved PSTP
- the full laboratory testing results report must be sent to the GRA for review
- a description of the materials used within, and to manufacture, the device must be provided to the GRA, along with information and certifications attesting to the safety of the materials for contact with drinking water. This includes information on any active agents used as part of the treatment process. The GRA will review the materials for toxicity and existing safety certifications. If the materials are not certified as safe for contact with drinking water, and the GRA toxicological review cannot conclude they are safe, extraction/leaching testing IAW the applicable NSF standard may be required. The test sponsor is responsible for scheduling and funding any required extraction/leaching testing.

If these steps are followed, the GRA will provide the test sponsor with a letter stating whether the results and material safety check warrant device compliance with NSF P248.02. If compliance is not granted, an explanation of why will be provided. In all cases, compliance statements will only cover the chemicals tested, as no blanket P248.02 compliance is available.

7.1.4 The DHA-PH, Environmental Health Sciences and Engineering Directorate/Field Water Branch may be contacted at dha.apg.pub-health-a.list.ehed@health.mil for clarification.

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Normative Annex 1

Suggested outline for a product-specific test plan

Title page

Executive summary

The Executive Summary describes the contents of the PSTP (not to exceed two pages). A general description of the equipment shall be included, as well as the testing location, a schedule, and a list of participants.

Table of contents

1. Purpose

2. Small water purifier capabilities and description

The PSTP shall include:

- description of the candidate SWP system, including all component parts and functions, to include photographs of the physical system from relevant angles or perspectives. The description of the equipment shall include at least the following information:
 - equipment name
 - model number
 - manufacturer name, address, and phone number
 - principles of operation and chemical contaminant reduction
 - size and weight
 - estimated production capacity and cleaning procedures
 - logistical requirements.
- an introduction and discussion of the engineering and scientific concepts on which the chemical reduction capabilities of the candidate SWP are based
- a description of the equipment and each process included as a component in the modular system including all relevant schematics
- a brief description of the physical construction/components of the SWP, including the general notable limitations, required consumables, ruggedness, etc.
- a statement of typical rates of consumption of chemicals.

3. ETO name and credentials

4. Device testing procedures

The PSTP shall include the following:

- detailed conditioning and operating procedures for the SWP with listing of operating parameters

- test water characteristics, application, and frequency of analysis of waters for ensuring compliance with required water characteristics
- chemical contaminant challenge, processing, and analytical procedures
- sampling and analysis schedule for water quality and chemical contaminant parameters
- requirements for initiating device cleaning and test completion
- procedures for determination of effective use for chemical contaminant reduction for the test capacity
- description of the statistical design, QA/QC procedures and statistical techniques that will be used to analyze the data
- identification of additional parameters of treated water quality and analytical methods that will be employed
- description of the methodologies to be used for conducting the chemical contaminant reduction studies
- outline of the testing schedule.

5. Roles and responsibilities

The PSTP shall include roles and responsibilities of all involved with testing. This shall specifically address the test sponsor and their relationship with the ETO as described below.

The test sponsor shall be responsible for the following:

- provision of new, commercially packaged equipment for testing complete with instructions
- provision of logistical and technical assistance to the ETO, as required
- provision of a complete list of all equipment and materials to be used in the verification testing
- provision of field operating procedures.

The ETO is responsible for the following:

- assisting the test sponsor in developing the PSTP and conducting efficacy testing IAW the PSTP
- providing needed logistical support
- establishing a communication network
- scheduling and coordinating the activities of all testing participants
- managing, evaluating, interpreting, and reporting on all test results; and evaluating and reporting on the performance of the SWP.

The ETO shall further be responsible for including the following elements in the PSTP:

- a definition of the roles and responsibilities of appropriate medical efficacy testing
- a table which includes the name, affiliation, and mailing address of each participant, a POC, description of participant's role, telephone and fax numbers, and email addresses
- a schematic of the organizational structure of operational and analytical support
- the GRA shall be responsible for clarifying this protocol to test sponsors and ETOs as required, and for reviewing and approving the PSTP in a timely manner.

Appendix A – References**Appendix B – Abbreviations and acronyms****Appendix C – Quality assurance project plan**

The responsibility for ensuring that SWP operation, sampling, and analysis activities comply with the QA/QC requirements of the PSTP shall rest with the ETO. QA/QC activities for a laboratory that analyzes samples sent off-site shall be the responsibility of that analytical laboratory's supervisor. The PSTP shall include:

- description of methodology for measurement of accuracy and precision
- description of the methodology for use of blanks, the materials used, and the frequency
- criteria for acceptable method blanks and the actions if criteria are not met
- definition of data to be reported during the Testing Program, in terms of analytical parameter type and frequency
- description of all techniques to establish (where applicable) the representativeness completeness, accuracy, and precision of methods in the analysis of water quality parameters and chemical contaminants.

Appendix D – Health and safety plan

The PSTP shall address safety considerations that are appropriate for the equipment being tested and for the chemical contaminants, if any, being used in the verification testing.

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